QPCODE

Reg.No : .....
Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) EXAMINATION MONTH 2025
FIFTH SEMESTER
DSE- MG5DSEBIF300- EVOLUTIONARY BIOLOGY AND MOLECULAR PHYLOGENETICS
(2024 ADMISSION ONWARDS)

Duration: 2hrs **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Questions

Answer any **5** questions

Each question carries **2** marks

1.	Discuss MSA.	[U] / [CO2]
2.	Describe UPGMA.	[A] / [CO3]
3.	Differentiate between rooted and unrooted tree.	[U] / [CO1]
4.	Relate cladogram.	[U] / [CO2]
5.	Indicate the difference between MP and ML tree.	[U] / [CO4]

6.	Show different softwares used for tree construction.	[A] / [CO5]
7.	Produce any two applications of phylogenetics in Medicine.	[A] / [CO6]

[2x5 = 10]

Part B

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

8.	Discuss the theory of molecular evolution.	[U] / [CO1]
9.	Differentiate between cladogram and phylogram.	[U] / [CO2]
10.	Compute distance based methods.	[A] / [CO3]
11.	Describe advantages and disadvantages of different tree building methods.	[U] / [CO4]
12.	Compile the applications of phylogenetic in agriculture.	[A] / [CO6]
13.	Illustrate the maximum parsimony method.	[A] / [CO3]
14.	Use any three tests for estimating phylogenetic tree.	[A] / [CO3]
15.	Write on convergent and divergent evolution.	[U] / [CO1]

[5x6 = 30]

Part C

Long Essay Questions

Answer any **3** questions

Each question carries **10** marks

14.	Discuss the different steps to construct a phylogenetic tree.	[U] / [CO2]
15.	Summarize the basic terminologies of a phylogenetic tree.	[U] / [CO1]
16.	Produce the basic applications of phylogenetics in agriculture and medicine.	[A] / [CO6]
17.	Summarize the different types of character based methods.	[U] / [CO4]
18.	Discuss the Darwin theory of evolution.	[U] / [CO1]

[10x3 = 30]